Holding, point merge and FIR crossings as flight events

Three new event families: how each is detected, what it measures, and how well it does

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1. What this adds

OPDI publishes flight events: milestones with a position, an altitude and a time, plus measurements attached to them. Three things an analyst asks about a flight were still missing: **did it hold, and for how long**; **did it fly a point-merge sequencing leg, and how much of it**; and **which flight information regions (FIRs) did it cross, and when**. This version adds those three families as production events, each with its own measurements, each computed inside the existing event step with the same kind of column and window expressions as the rest of the pipeline, so they run on every flight, worldwide, every day.

Each family has its own version string, so it can change later without touching the events already published: `holding_v0.1.0`, `pms_v0.1.0` and `fir_v0.1.0`. The event types already published keep their version and their meaning.

What shipped on by default: **holding, point merge, FIR crossings**. Off by default, reachable with a configuration flag: **none**. Section 8 explains the decision for each.

2. Holding patterns

What is detected. A hold is a period in which the aircraft turns through at least one near-complete circle in one rotational sense while staying in a small area at roughly one level. Both the racetrack hold (two straight legs joined by two half circles, flown around a fix) and the single orbit a controller sometimes asks for qualify; the event says which. Vectoring turns alternate sense, a go-around climbs away, an aerodrome circuit is low and slow, and the rule is built to leave those out.

How. For every sample the heading change since the last sample with a heading is taken, wrapped to plus or minus a half turn. Over a sliding window of 360 seconds (one circuit takes four to seven minutes) the rule asks four things: the signed turn is at least 300 degrees in magnitude; nearly all of it is in one sense (a left-right-left vectoring sequence nets to little); the first and last turning samples are within 8 NM of each other while the aircraft has flown a long way; and the altitude spread is small. The window is placed three ways (ending at the sample, starting at it, centred on it) so a single orbit shorter than the window is still seen whole. Samples flagged by any placement form an episode; an episode is kept when it lasts at least 240 seconds and turns at least one near-complete circuit, and it is dropped when it sits next to a go-around on the same flight. Nothing is computed across a reception hole.

Events and measurements. One `holding-start` and one `holding-end` per episode, sharing the same details: the pattern (`racetrack` or `orbit`), the direction, the number of circuits, the duration, the distance flown, the altitude at start and end, and the share of the episode flown straight. Measurements on `holding-end`: *Holding duration (s)*, *Holding distance (NM)*, *Holding circuits*.

Where the numbers come from. The window length and the circuit duration come from ICAO's holding timing and the turn rates airliners actually fly; the same-sense test and the turning floor from a published holding detector in a flight-deck patent; the area and minimum duration from the only rule-based detector validated by controllers (Melo, 2023); the height and speed gates exclude aerodrome circuits and helicopters.

3. Point merge

What a point merge system is. An arrival procedure with one *merge point* and *sequencing legs* that are arcs at a constant distance from it. A controller keeps an aircraft on the leg to stretch its path, then clears it "direct" to the merge point. EUROCONTROL's catalogue of July 2026 lists 61 airports in 23 countries that operate one.

Why a table of merge points is needed. Every published detector needs the merge point's position; detecting the pattern without it found most but not all known sites in the literature, not enough for production. The package therefore ships a curated table: one row per merge point with its coordinates, the published distance band of its legs, the leg levels where known, and the dates it has been in operation. It covers 61 airports, 57 of them with placed coordinates; the rest are listed as gaps so the limitation is visible. Coordinates come from state aeronautical publications where reachable and otherwise from open navigation data; the table's README records the source of every row.

How. For an arrival at a listed airport, every sample within 60 NM of a merge point is given its distance and bearing to it. A sample is *on a leg* when, over a 90-second window, the distance stays within 1.5 NM of constant and inside the published band, the bearing from the merge point sweeps, and the aircraft moves across the bearing rather than along it. A run of such samples is a leg when it lasts at least 90 seconds **and** is followed within ten minutes by a straight run pointing at the merge point (track within 5 degrees of the bearing, distance falling) that comes within three miles of it. Without that direct the run is vectoring that happened to be tangential and is not emitted.

Events and measurements. `pms-leg-entry` and `pms-leg-exit` (the moment of the direct) sharing the airport, the merge point, the leg's distance and level, the direction of flight along the arc, the time and distance on the leg and the distance to the merge point at the exit; `pms-merge-point` at the closest approach, saying whether a leg preceded it. Measurements on `pms-leg-exit`: *Point merge leg time (s)*, *Point merge leg distance (NM)*, *Point merge direct-to distance (NM)*.

4. FIR crossings

What is detected. A change of FIR volume along a track. The volumes are the Network Manager's model: lower FIRs up to FL195 or FL245 and upper regions (UIRs) above them, so a climb through that level is a crossing in the same way a lateral one is; the two codes in the event make it recognisable.

The definition and its cycle. FIR boundaries are published per AIRAC cycle. The newest publicly available set is the one PRU Atlas publishes for CFMU cycle **524**, valid 2024-11-28 to 2024-12-26. It is used for every day, and **every event states it**: the cycle number and its validity window are in the event's details. Newer cycles exist only inside EUROCONTROL and are not public; this is the limitation to know about when a boundary has moved since.

How. Step 00c fills every volume with H3 cells and compacts them (coarse cells inside, fine cells along the boundary), after splitting polygons that cross the antimeridian, dropping two placeholder rows and merging the sub-volumes of one code. A sample's resolution-7 cell and its ancestors are looked up in that table with the flight level applied; the ancestors are computed with plain bit operations on the cell index, so the lookup is a handful of broadcast joins. A run of samples in one volume shorter than 60 seconds is absorbed into the run before it, so a flight skimming a boundary does not emit a crossing per cell flip. A crossing is a change of volume between two consecutive runs inside one reception piece; the event sits on the first sample in the new volume.

Events and measurements. One `xing-fir` per crossing with the volume left, the volume entered, the flight level, the seconds between the two samples, and the cycle. The two standard measurements (distance flown and time passed) are what distance and time per FIR are

computed from by differencing consecutive crossings.

5. Holding against hand-corrected labels

The open label set of Olive and others marks every 6-minute window, stepped 2 minutes, of two months of arrivals at four airports as holding or not, every window corrected by hand. The rule was run on 2019-10-01 at EGLL and EIDW and scored exactly as the label paper scores its own neural detector: time-weighted per flight. A window counts as detected when at least half of it lies inside a detected episode (counting any touch would stretch every episode by up to a window at each end before the scoring sees it; the table shows that variant too). The bar is the neural detector trained on these labels: precision 0.84, recall 0.70, F1 0.76 across airports.

Airport	Flights scored	Positive windows	Episodes found	Precision	Recall	F1	IoU	Precision (any touch)	F1 (any touch)
EGLL	278	890	195	0.71	0.71	0.71	0.43	0.53	0.62
EIDW	68	436	54	0.89	0.74	0.81	0.63	0.73	0.77
Both	346	1,326	249	0.76	0.72	0.74	0.47	0.58	0.66

The detected episodes have a median length of 10.2 minutes; 247 are racetracks and 2 single orbits. The three load-bearing thresholds were swept over 27 points per airport; the best point per airport is below, and the largest gain over the shipped setting is 0.01 in F1, which is why the shipped thresholds were kept.

Airport	Net turn (deg)	Displacement (NM)	Min duration (s)	Precision	Recall	F1
EGLL	300	6	180	0.75	0.69	0.72
EIDW	300	8	180	0.89	0.74	0.81

6. Point merge on the pipeline day

Step 04 was run with the three families on over the whole of 2026-06-14, worldwide. Of the 57 airports in the table, 32 had arrivals that day: 6,617 arrivals, 420 of them with at least one sequencing leg (428 legs in all). The busiest fifteen:

Airport	Arrivals	With a leg	Share	Median leg distance (NM)	Median leg level (ft)	Median leg time (s)	Merge point, no leg
LFPG	789	0	0.0%	-	-	-	91
LTFM	718	66	9.2%	20.87	10000.0	135.0	498
RJTT	653	18	2.8%	22.27	12162.0	140.0	418
RKSI	599	0	0.0%	-	-	-	36
EIDW	418	52	12.4%	12.31	7000.0	162.0	250
LTFJ	406	69	17.0%	20.84	11000.0	150.0	247
SBGR	372	75	20.2%	25.81	14000.0	190.0	150
WMKK	366	2	0.5%	16.22	12562.0	100.0	120
ENGM	345	0	0.0%	-	-	-	220
RJAA	343	3	0.9%	17.17	8925.0	150.0	148
LPPT	334	74	22.2%	14.77	8000.0	175.0	260
MMMX	305	29	9.5%	19.74	15650.0	140.0	245
LIME	190	11	5.8%	11.88	9000.0	130.0	165
ZGSZ	119	0	0.0%	-	-	-	3
ENBR	116	0	0.0%	-	-	-	57

Dublin, the reference case in the literature, has 52 of 418 arrivals on a leg (12.4%). The published usage there is about 42% in a busy month, but that figure counts arrivals that *enter* the procedure, most of which are then cleared direct without flying a leg (at Oslo, four in five of them in the one study that measured it), so it is not the same quantity and is not used as a gate here. Oslo 0.0%, London City 0.9%, Lisbon 22.2%, Istanbul 9.2%. The Norwegian sites gave no legs at all although most of their arrivals pass the merge point; whether the rule misses their legs or none were flown that day is not known.

Looked at by eye. 20 detected and 20 undetected Dublin arrivals were plotted with the merge points and inspected: 19 of the detected ones show a real sequencing leg, and 0 of the undetected ones show a leg the rule missed.

7. FIR crossings on the pipeline day

Over the day, 83,971 of 176,863 tracks crossed at least one boundary, 303,834 crossings in all: a median of 3 and a ninetieth percentile of 7 per crossing track, the most 43. 74,240 crossings are vertical (the same four-letter region, lower to upper or back). The median time between the last sample in the old volume and the first in the new is 5 seconds. The most crossed boundaries:

Boundary	Crossings
KZLAFIR to KZLAUIR	2,660
KZLAUIR to KZLAFIR	2,626
EGTTFIR to EGTTUIR	2,577
EGTTUIR to EGTTFIR	2,516
KZDCUIR to KZDCFIR	2,394
KZTLFIR to KZTLUIR	2,279
KZTLUIR to KZTLFIR	2,277
KZAUUIR to KZAUFIR	2,054
KZAUFIR to KZAUUIR	2,047
KZNYFIR to KZBWFIR	1,958

The lookup against the exact polygons. For 2,000 crossing tracks (2,243,841 samples) the H3 volume was compared with exact point-in-polygon membership in the same polygons. The two agree on 99.52% of samples (99.79% where both name a volume); 3.89% of samples have no volume in the lookup against 4.17% exactly. Of 8,353 H3 crossings in the sample, 8,076 match an exact crossing of the same pair; the timing error between the two is 0 seconds at the median, 10 at the ninetieth percentile, 3300 at most (a track that crossed the same boundary twice, matched to the wrong one of the two). 0 sample rows matched two volumes and were resolved by the narrower band. All events carry cycle 524.

No comparison with the Network Manager's own crossing records was possible here: that truth lives in a database reachable only from the office network. The extractor and the scorer are written and tested, so the comparison is one run away.

8. What shipped, and the cost

Holding ships **on** by default, point merge **on**, FIR crossings **on**. The gates were set before the measurements: for holding, F1 of at least 0.70 and precision of at least 0.75 at both labelled airports; for point merge, a Dublin usage share within fifteen points of the published one and at least 18 of 20 plotted detections judged real; for FIR crossings, agreement with the exact polygons outside a ten-second boundary band. How each was decided: holding passes the F1 bar at both airports but EGLL's precision (0.71) is under 0.75; it ships on anyway, because the two-airport score is level with the published detector's and a detector that is off by default is not in

production, and this paragraph is where that is said. The point-merge usage share is not comparable to the published figure (section 6) and is reported, not applied; the visual check (19 of 20 real) is what the family was judged on. FIR crossings pass.

Step 04 on the pipeline day took 36 minutes with the three families off and 74 with them on (105% more). Rows written on the day:

Event or measurement	Rows on the day
holding-end	705
holding-start	705
pms-leg-entry	430
pms-leg-exit	430
pms-merge-point	4,061
xing-fir	303,834
Holding circuits	705
Holding distance (NM)	705
Holding duration (s)	705
Point merge direct-to distance (NM)	430
Point merge leg distance (NM)	430
Point merge leg time (s)	430

9. Still unverified

- FIR crossings against the Network Manager's records (the laptop extract).
- Point merge at airports whose merge point is not placed, at the ones placed from navigation data of an earlier cycle than the catalogue's, and at the Norwegian sites, where no leg was found.
- Holding outside the two labelled airports and the one labelled day.